NHS Elective Capacity: Analysis, Optimisation and Scenario Strategy

Self-directed end-to-end engagement for NHS Cheshire and Merseyside ICB (QYG). No client, no brief, real NHS England open data.

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github.com/Vedant-Au/nhs-elective-capacity-analysis
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THE QUESTION

An ICB has a constrained elective recovery budget, three ways to spend it, and a backlog it cannot clear. Where should the money go, and what would tell us the answer was wrong?

1 The evidence base had to be built first

No dataset existed for the question. Seven NHS England sources across 84 months were warehoused in DuckDB, then a pressure index, Monte Carlo forecasting, clustering and an inequality-of-access layer were built on top, followed by a linear-programming capacity optimiser and a scenario model.

2 It is a money problem, not a capacity problem

No delivery lever's capacity ceiling binds until funding reaches £31.9m for independent sector, £69.7m for diagnostic or £107.1m in-house. The range under consideration is £7.5m to £30m, so three long-running deliverability debates change the outcome by nothing at all.

3 The entire allocation turns on one unpublished number

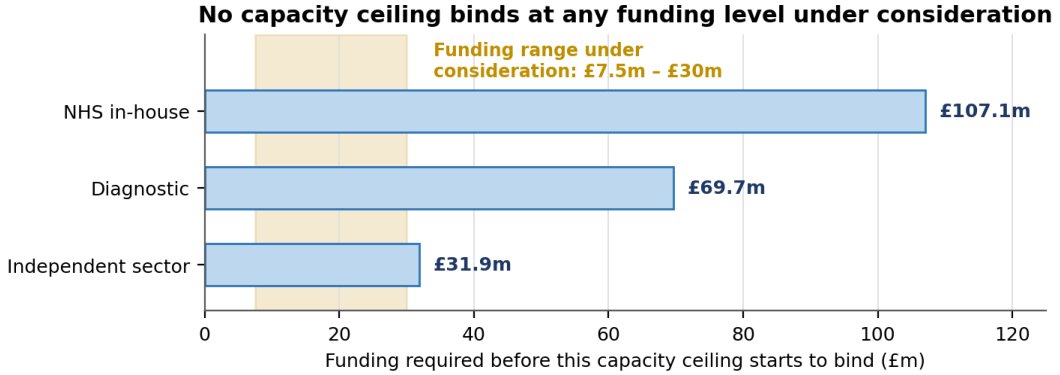
Diagnostic and treatment capacity become indifferent at 3.34 diagnostic tests per unlocked pathway. The planning assumption is 1.75 and no NHS dataset publishes the figure. Triangulating against observed DM01 activity gives 0.84, which bounds the argument rather than settling it.

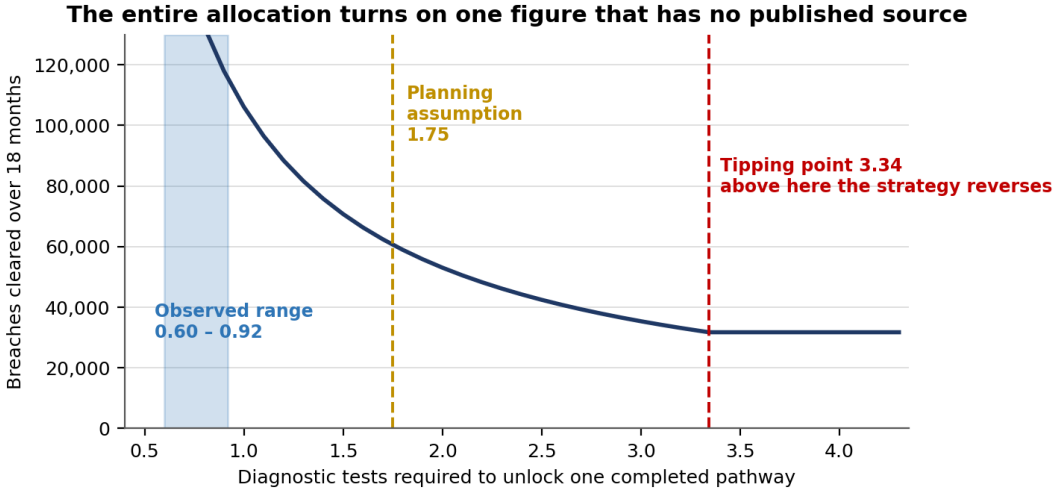
4 Equity is free, and one expected finding did not survive

At every funding level examined, the whole reduction can go to higher-deprivation trusts at zero cost to total breaches cleared. Separately, more-deprived catchments turned out to have *more* consultants per 1,000 FTE, not fewer (Spearman rho –0.59, p = 0.044).

ASSURANCE, AND WHY THE RECOMMENDATION IS A JUDGEMENT

Validation caught three defects, all documented rather than quietly fixed. One was found only by diffing the generalised toolkit against the engagement engine, one cell in forty, and it reversed the recommendation: before the fix the answer looked robust under every weighting, after it the choice depends entirely on risk appetite. A statistical claim was also retracted as endpoint selection. The rule now enforced throughout: never rank a quantity the optimisation does not pin.





Built with DuckDB · Python · Excel Solver · Tableau · linear programming, Monte Carlo, scenario modelling

Delivered warehouse · analytics · optimiser · Tableau story · board paper · reusable toolkit